

1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: _____

June 25, 2009 Thursday

Department/School: _____

SID: _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

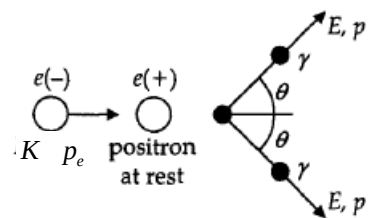
1.(20 pts.) A rod of length L_0 moves with a speed v along the x -direction. The rod makes an angle θ_0 with the x' -axis.

(a) Show that the length of the rod as measured by a stationary observer is given by

$$L = L_0 \left[1 - (v^2/c^2) \cos^2 \theta_0 \right]^{1/2}.$$

(b) Show that the angle that the rod makes with the x -axis is $\theta = \tan^{-1}(\gamma \tan \theta_0)$ where $\gamma = 1/\sqrt{1-v^2/c^2}$.

2.(20 pts.) An electron having kinetic energy $K = 1.000 \text{ MeV}$ makes a head on collision with a positron at rest. In the collision the two particles annihilate each other into two γ -ray photons of equal energy, each traveling at equal angles with the electron's direction of motion, as shown in the figure. Find the energy E , momentum p and angle of emission of the γ -rays. (The electron rest mass is $0.511 \text{ MeV}/c^2$.)



3.(20 pts.) A mole of the ideal monatomic gas occupies a volume V_i at a pressure p_i . The gas undergoes a process in which the pressure is proportional to the volume. At the end of the process, it is found that the root-mean-square speed v_{rms} of the gas molecules has doubled from its initial value. Determine the amount of energy transferred to the gas by heat.

4.(20 pts.) The process $\text{H}_2\text{O} (\text{liquid}) (-10^\circ\text{C}) \rightarrow \text{H}_2\text{O} (\text{solid}) (-10^\circ\text{C})$ of 1 mol of H_2O is carried out in an isolated system consisting of the 1 mol of H_2O and a thermostat at constant temperature of -10°C . The latent heat for melting ice is 333J/g . The specific heat capacity of water is $4.2\text{J/g}^\circ\text{C}$ and the specific heat capacity of ice is $2.1\text{J/g}^\circ\text{C}$.

(a) Calculate the change in entropy of the 1 mol of H_2O in this process.

(b) Calculate the change in entropy of the thermostat.

(c) Is this process spontaneous? Why?

(Hint: The process of freezing of the super cooled water is irreversible. Thus, to calculate the entropy change in the process $\text{H}_2\text{O} (\text{liquid}) (-10^\circ\text{C}) (\text{state 1}) \rightarrow \text{H}_2\text{O} (\text{solid}) (-10^\circ\text{C}) (\text{state 2})$, one has to find a reversible path connecting the state 1 and state 2.)

5.(20 pts.) Assume that the internal energy of a mole of gas that obeys the Van der Waals equation of state

$$\left(p + \frac{a}{V^2} \right) (V - b) = RT$$

is given by

$$U = U_0 + C_V T - \frac{a}{V}$$

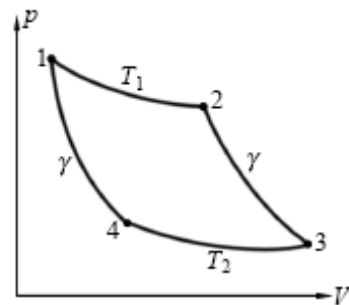
where C_V is considered to be a constant. The gas undergoes

a Carnot cycle between two heat reservoirs of temperature T_1 and T_2 , as shown in the accompanying figure.

(a) Show that the adiabatic equation of the gas is given by $T(V - b)^{R/C_V} = C$.

(b) Calculate the work done by the system and the heat input for the cycle.

(c) Obtain the thermal efficiency for the cycle in terms of T_1 and T_2 .



6.(20 pts.) A mole of gas obeys equations

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{R}{p} + \frac{a}{T^2}, \quad \left(\frac{\partial V}{\partial p}\right)_T = -Tf(p)$$

where a is a constant and $f(p)$ is a function of pressure p only.

(a) Find $f(p)$.

(b) Find the equation of state of the system.

(c) The constant-pressure (isobaric) heat capacity of the system can be expressed as $C_p = T\left(\frac{\partial S}{\partial T}\right)_p$.

Show that $\left(\frac{\partial C_p}{\partial p}\right)_T = \frac{2a}{T^2}$.